

Algoritmos de CYK - Introdução à Teoria

1. Dada as seguintes gramáticas abaixo, utilize o algoritmo de reconhecimento e mostre se as cadeias não ou não são geradas por uma gramática.

La (2 SA3, 2 W, 3 P, 5)

P = 2 S - AAIA5126

A → SAIASI v. 3

$$m = 1000$$
[illegible]

AAB

B A A B

$$A \cdot AB = A \quad SA \rightarrow AS \quad AA$$
$$B A A B = S \quad S A \rightarrow \begin{matrix} & X & A \\ S S & & S A \end{matrix}$$
$$AA \ B = S \ S \rightarrow S^X S$$
$$BA \cdot AB = A \quad \begin{matrix} SA & S \\ AS & AA \end{matrix} \rightarrow 0$$
$$BAA \ B = 3 \ 5 \rightarrow 55$$

ABAA

$$A \text{ BAA} = A \text{ S} \rightarrow \begin{matrix} \text{S,A} \\ \text{AS} \end{matrix}$$

ABAAAB

$$AB \quad AA = SA \quad S \rightarrow SS \quad AS$$
$$A \text{ BAAB} = A \text{ SA} \rightarrow \begin{matrix} \text{SA} & \text{S} \\ \text{AS} & \text{AA} \end{matrix}$$
$$ABA \quad A = SA \quad A \rightarrow SA \quad A$$
$$AB \ AAB = SA \ SA \rightarrow \begin{matrix} & & X & & A & SA & S \\ & & SS & & SA & AS & AA \end{matrix}$$
$$ABA \quad AB = SA \quad SA \rightarrow SS \quad SA \quad SA \quad S$$

Linie gerade

$$ABAA \quad B = SA \quad S \rightarrow SS \quad \begin{matrix} X & SA \\ SS & AS \end{matrix}$$

0.5-yy1y510

y-5y1y511

011011

6	S,y						
5	S,y	S,y					
4	S,y	S,y	S,y				
3	y	S,y	S,y	S			
2	y	S	S,y	y	S		
1	S	y	y	S	y	y	
	0	1	1	0	1	1	

011

0 11 -> S S = S^x

0 1 1 -> y y = y^s

110

1 10 -> y S y = y^{S,y} y^S

1 1 0 -> S S = S^S

101

1 01 -> y y = y^S

1 0 1 -> S y y = S^S y^S

1011

1 011 -> y y = y^S

1 0 11 -> S y S y = S^S S^S y^S y^S

1 0 1 1 -> S y y = S^S y^S

011

0 11 -> S S = S^x

1 1 0 -> y y = y^S

0110

0 110 -> S S y = S^x S^y

0 1 10 -> y S y = y^{S,y} y^S

0 1 1 0 -> y S = y^S

01101

0 1101 -> S S y = S^x S^y

0 1 101 -> y S y = y^{S,y} y^S

0 1 1 01 -> y y = y^S

0 1 1 0 1 -> S y y = S^S y^S

1101

1 101 -> y S y = y^{S,y} y^S

1 1 01 -> S y = S^S

1 1 0 1 -> S y y = S^S y^S

11011

1 1011 -> y S y = y^{S,y} y^S

1 1 011 -> S y = S^S

1 1 0 11 -> S y S = S^S S^S y^S

1 1 0 1 1 -> S y y = S^S y^S

no operados

011011

$$011011 \rightarrow S \text{ sy} = \overset{x}{SS} \overset{y}{sy}$$

$$011011 \rightarrow y \text{ sy} = \overset{sy}{ys} \overset{y}{yy}$$

$$011011 \rightarrow y \text{ y} = \overset{x}{yy}$$

$$011011 \rightarrow sy \text{ s} = \overset{x}{ss} \overset{sy}{sy}$$

$$011011 \rightarrow sy \text{ y} = \overset{y}{sy} \overset{s}{yy}$$

100100

6	Sy						
5	Sy	Sy					
4	Sy	Sy	Sy				
3	Sy	y	Sy	y			
2	Sy	-	y	Sy	-		
1	y	s	s	y	s	s	
	1	0	0	1	0	0	

100

$$100 \rightarrow y - = \overset{x}{y}$$

$$100 \rightarrow sy \text{ s} = \overset{x}{ss} \overset{sy}{sy}$$

001

$$001 \rightarrow s \text{ y} = \overset{y}{sy}$$

$$001 \rightarrow - \text{ y} = \overset{x}{y}$$

0010

$$0010 \rightarrow s \text{ sy} = \overset{x}{ss} \overset{y}{sy}$$

$$0010 \rightarrow - \text{ sy} = \overset{y}{sy}$$

$$0010 \rightarrow y \text{ s} = \overset{sy}{ys}$$

010

$$010 \rightarrow s \text{ sy} = \overset{x}{ss} \overset{y}{sy}$$

$$010 \rightarrow y \text{ s} = \overset{sy}{ys}$$

100

$$100 \rightarrow y - = \overset{x}{y}$$

$$100 \rightarrow sy \text{ s} = \overset{x}{ss} \overset{sy}{sy}$$

0100

$$0100 \rightarrow s \text{ sy} = \overset{x}{ss} \overset{y}{sy}$$

$$0100 \rightarrow y - = \overset{x}{y}$$

$$0100 \rightarrow sy \text{ s} = \overset{x}{ss} \overset{sy}{sy}$$

1001

$$1001 \rightarrow y \text{ y} = \overset{s}{yy}$$

$$1001 \rightarrow sy \text{ y} = \overset{s}{sy} \overset{s}{yy}$$

$$1001 \rightarrow sy \text{ y} = \overset{s}{sy} \overset{s}{yy}$$

10010

$$10010 \rightarrow y \text{ sy} = \overset{sy}{ys} \overset{s}{yy}$$

$$10010 \rightarrow sy \text{ sy} = \overset{ss}{ss} \overset{sy}{sy} \overset{sy}{sy} \overset{s}{yy}$$

$$10010 \rightarrow sy \text{ sy} = \overset{x}{ss} \overset{sy}{sy} \overset{sy}{sy} \overset{s}{yy}$$

$$10010 \rightarrow sy \text{ s} = \overset{x}{ss} \overset{sy}{sy}$$

00100

$$0 \ 0100 \rightarrow S \ 5y = \overset{x}{55} \overset{y}{5y}$$

$$00 \ 100 \rightarrow - \ 5y = \overset{y}{5y}$$

$$001 \ 00 \rightarrow y - = \overset{x}{y} \overset{y}{5y}$$

$$0010 \ 0 \rightarrow 5y \ 5 = 55 \ y5$$

100100

$$1 \ 00100 \rightarrow y \ 5y = \overset{5y}{y5} \overset{5}{y5}$$

$$10 \ 0100 \rightarrow 5y \ 5y = 55 \ 5y \ y5 \ y5$$

$$100 \ 100 \rightarrow y \ y = \overset{5}{y5}$$

$$1001 \ 00 \rightarrow 5y - = \overset{x}{5y}$$

$$10010 \ 0 \rightarrow 5y \ 5 = 55 \ y5$$